5 Waves can be transverse or longitudinal.

(a) (i) Complete the sentence to describe a longitudinal wave.

In a longitudinal wave, the direction of vibration is ..... to the direction of propagation. [1]

(ii) Give **one** example of a longitudinal wave.

..... [1]

(b) A vibrating rod produces a water wave in a ripple tank.

(i) The wavelength of the wave is 0.016 m.

The wave is travelling at a speed of 0.12 m/s.

Calculate the frequency of the wave. Give the unit.

frequency = ... unit ..... [2]

(ii) The water wave travels from shallow water to deep water.

Complete Table 5.1 to show what happens to the properties of the water wave.

Indicate your answers by placing ticks (✓) in the appropriate boxes. One has been done for you.

Table 5.1

	from shallow water to deep water		
wave property	**decreases**	**stays the same**	**increases**
wavelength			✓
frequency			
speed			

[2]

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(c) Figure 5.1 shows a ripple tank demonstrating a wave effect.

The wave passes through a small gap in a barrier.

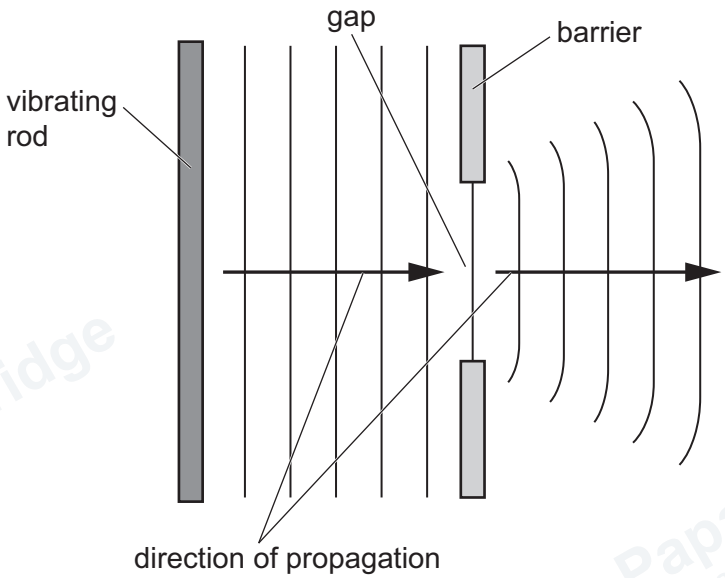

Figure 5.1

(i) State the name of the wave effect shown in Figure 5.1.

..... [1]

(ii) State **two** changes that each increase the curvature of the wavefronts as the wave passes through the gap.

1 .....

2 .....

[2]

[Total: 9]

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